RAPPORT FOR LASERSKANNING

Geovekstprosjekt

LACHNO43_45 Nordland 2014 – 2.0pkt

terratec

INNHOLDSFORTEGNELSE

1. GENERELLE OPPLYSNINGER I PROSJEKTET 3

1.1 OPPDRAGSGIVER 3
1.2 OPPDRAGET 3
1.3 OPPDRAGSTAKER 3
1.4 OPPDRAGETS INNHOLD 3
1.5 KVALITETSSIKRING 3
1.6 KARTUTSNITT 4

2. DATAINNSAMLING 5

2.1 UTFØRELSE 5
2.2 MÅLEUTSTYR 5
2.3 GRUNNLAG/DATUM 5
2.4 KVALITET PÅ KONTROLLPUNKT 5
2.5 AVVIK KONTROLLPUNKT 5
2.6 TRANSFORMASJONER 6
2.7 VANSKELIGHETER I DATAINNSAMLINGEN 6

3. PROSESSERING/DATABEARBEIDING 6

3.1 BEREGNINGS- OG EDITERINGSARBEID 6
3.2 GNSS/INS-BEREGNINGER 6
3.3 KALIBRERING OG STRIPEJUSTERING 6
3.3.1 Prosjektkalibrering 7
3.4 RESULTAT FRA KJENTFLATER 11

4. LEVERANSER 12

4.1 INNHOLD I LEVERANSENE 12
4.2 OPPDELING OG NAVNING AV FILER 12
5.0 VEDLEGG A 13

Rapport utarbeidet,

Vækerø, 05.01.2015

Olav Ytterdal
Prosjektleder

1. GENERELLE OPPLYSNINGER I PROSJEKTET

1.1 Oppdragsgiver

Navn: Kartverket Bodø
Besøksadresse: Statens hus, Molovn. 10, Bodø
Postadresse: Molovn. 10, 8002 Bodø
Prosjektleder: Christian Malmquist

1.2 Oppdraget

Navn:	LACHNO43_45
Kommune, -nr:	Bodø (1804), Narvik (1805), Bindal (1811), Alstadhaug (1820), Leirfjord (1822), Vefsn (1824), Træna (1835), Meløy (1837), Gildeskål (1838), Saltdal (1840), Fauske (1841), Sørfold (1845), Steigen (1848), Hamarøy (1849), Tysfjord (1850), Tjeldsund (1852), Evenes (1853), Flakstad (1859), Vestvågøy (1860), Vågan (1865), Hadsel (1866), Øksnes (1868), Sortland (1870)
Fylke:	Nordland

1.3 Oppdragstaker

Navn:	TerraTec AS
Besøksadresse:	Vækerøveien 3, 0281 Oslo
Postadresse:	Vækerøveien 3, 0281 Oslo
Prosjektleder:	Berit Morstad
Prosjektreferanse:	40445_Laser Nordland 2.0pkt

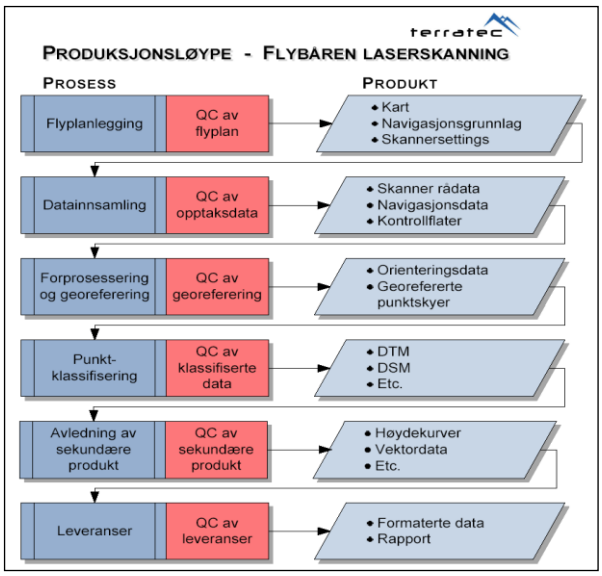
1.4 Oppdragets innhold

Oppdraget innebærer laserskanning av område med minimum punktetthet 2.0 pkt/m2. Det utføres preprosessering og sensor-/stripekorreksjoner for endelig georeferering av punktskyen. Absolutt høydenøyaktighet forbedres ved måling mot kontrollflater. For generering av terrengmodell utføres automatisk og manuell klassifisering av datasettet.

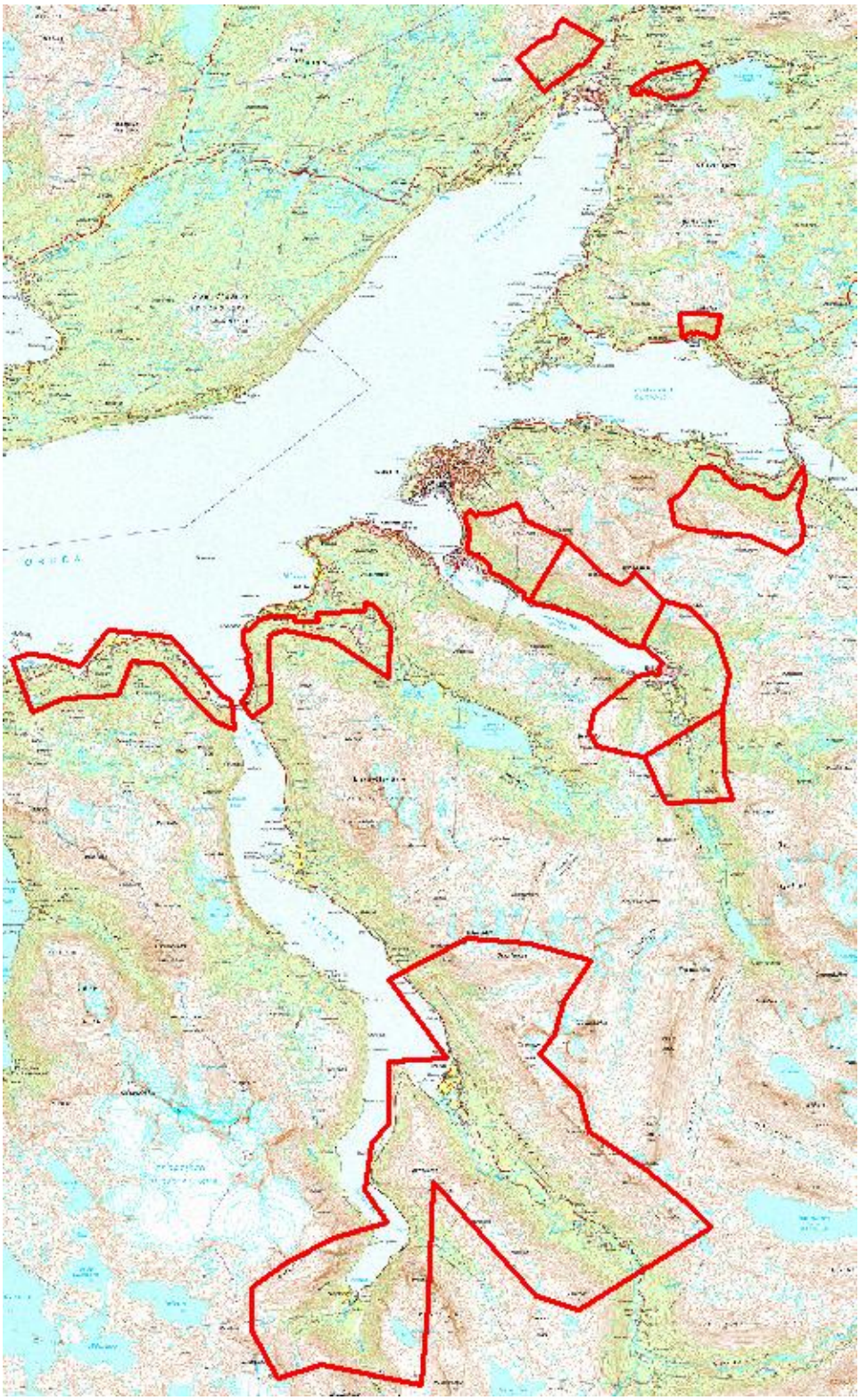
1.5 Kvalitetssikring

Kvalitetssikringsopplegget er utført i.h.h.t. TerraTecs kvalitetsstyrings-system. Opplegget for denne type produksjon er vist skjematisk i figuren til høyre. I dette prosjektet er det spesielt lagt vekt på:

- Kalibrering av sensorsystem
- Tverrstriper og flystriper i forskjellige retninger
- Utjevning mellom overlappende striper
- Justering mot kontrollflater



1.6 Kartutsnitt



2. DATAINNSAMLING

2.1 Utførelse

TerraTec AS har gjennomført laserskanningen i følgende operasjoner:

Område	Operatør	Dato
Nordland 2.0pkt	Jaanus Juhos	29.06.2014
Nordland 2.0pkt	Mariann Saarend	25.08.2014
Nordland 2.0pkt	Mariann Saarend	26.08.2014
Nordland 2.0pkt	Mariann Saarend	28.08.2014
Nordland 2.0pkt	Tarvo Hiiemaa	15.09.2014
Nordland 2.0pkt	Jaanus Juhos	16.09.2014

2.2 Måleutstyr

Kalibreringsparametre for systemet er beregnet ved flyging over vårt standard kalibreringsområde i Hamar. Disse parametrene er beregnet i TerraMatch og implementert i ALS Post Processor for prosessering av rådata.

Spesifikasjoner laserskanning Nordland 4.0pkt:

Laserskanner, snr: ##)	ALS70, snr. 208
IMU:	MicroIRS
GNSS-mottakere:	NovAtel OEM4 og Topcon Legacy
Plattform:	FW
Høyde over laveste terreng (m)	1460m-2449m
Flyhastighet:	150kt
Laserpuls	191000 Hz
Åpningsvinkel (grad)	±20 grader

2.3 Grunnlag/datum

Horisontalt:	Euref89 UTM sone 33, NTM sone 17
Vertikalt:	NN1954

2.4 Kvalitet på kontrollpunkt

Følgende kontrollflatemålinger er utført:

Utført av:	TerraTec AS
Antall flater:	21
Målemetode:	CPOS
Utstyr:	Leica 1200 GPS-mottakere

2.5 Avvik kontrollpunkt

Avviket mellom kontrollflater og laserdata er relativt lite og godt innenfor kravet - jfr. kap 3.4.

2.6 Transformasjoner

GPS-beregning i TerraPOS er utført i WGS84. Transformasjon er utført for levering i de ulike datum prosjektet skal leveres i.

Transformasjon WGS84 – EU89 UTM33

Transformasjon EU89 UTM33 – NTM 17

Høydetransformasjon Ellipsoidisk – NN1954

Høydetranformasjonenen fra ellipsoidiske til ortometriske høyder er utført med geoidemodell generert i WSK Trans, Href2007b utgitt av Statens kartverk.

2.7 Vanskeligheter i datainnsamlingen

Det er ikke rapportert om noen vanskeligheter i forbindelse med datainnsamlingen.

3. PROSESSERING/DATABEARBEIDING

3.1 Beregnings- og editeringsarbeid

Program/versjon	Produsent	Funksjon
Leica Mission Pro	Leica Geosystems	Flyplanlegging
TerraPOS	TerraTec	GNSS-prosessering
IPAS Pro	Leica Geosystems	GNSS/INS-integrasjon
ALS Post - Processing ALSPP	Leica Geosystems	Beregning av "rå" punktskyer
WSKTrans	Statens kartverk	Geoide
Leica Geo Offce	Leica Geosystems	Landmålingsberegninger
TerraScan	Terrasolid	Prosessering, visning, editering og klassifisering av laser punktdata
TerraMatch	Terrasolid	Sensor- og stripekorreksjoner
TerraModeler	Terrasolid	Visualisering av terrengoverflate
MicroStation	Bentley	CAD-plattform for Terrasolid-program

3.2 GNSS/INS-beregninger

Blending av GPS og IMU-data er utført med kalman-filter i Leicas programvare IPAS-PRO v 15. Det er for øyeblikket ingen god utskriftmulighet for dokumentasjon av beregningen. Alle resultatene av beregningene har høy kvalitet og er innenfor spesifikasjonene til sensorene.

3.3 Kalibrering og stripejustering

Utover årlig kalibrering av sensoren er det gjennomført en egen prosjektkalibrering for å redusere eventuelle gjenværende feil i sensororienteringen. Arbeidet er utført med TerraMatch.

3.3.1 Prosjektkalibrering

Det er foretatt en prosjektkalibrering for hver enkelt flysesjon. Her er korreksjoner for Heading, Roll og Pitch estimert og deretter påført. Det har ikke blitt estimert unormale verdier under denne prosessen. Deretter inngår alle flystriper i endelig stripeutjevning, hvor det blir løst for delta Z og delta R:

```
Tie lines:
H:\04_Laser_Projects_2014\40445_Nordland_2pt\Lidardata\reports\40445_
TIL_HPRS.til
Trajectories:
H:\04_Laser_Projects_2014\40445_Nordland_2pt\trajectories\
Solution for individual strips
Combined solution for all scanners

Starting avg 3d mismat 0.04093
Starting avg xy mismat 0.07346
Starting avg z mismatc 0.04068

Final avg 3d mismatch: 0.02214
Final avg xy mismatch: 0.05774
Final avg z mismatch:  0.02191

Execution time: 54.9 sec
Number of iterations: 42

Flightline     Z shift R shift
1               +0.023 -0.0083
2               +0.157 -0.0081
3               -0.063 -0.0183
4               -0.014 -0.0037
6               +0.087 -0.0018
7               +0.065 +0.0051
8               +0.043 +0.0006
9               +0.134 -0.0060
10              +0.066 -0.0004
11              -0.047 -0.0010
12              +0.025 +0.0003
13              +0.080 +0.0042
14              +0.103 -0.0060
15              +0.066 -0.0086
16              +0.080 +0.0029
17              +0.012 -0.0058
```

18	+0.025	+0.0016
19	+0.082	-0.0042
20	-0.037	+0.0076
21	+0.071	-0.0015
22	+0.025	-0.0090
23	-0.040	-0.0019
24	-0.065	-0.0001
25	-0.116	-0.0069
26	+0.005	-0.0008
27	-0.068	+0.0060
28	-0.071	-0.0060
29	-0.043	-0.0071
30	+0.007	+0.0020
31	-0.079	+0.0113
32	-0.057	-0.0159
33	-0.071	-0.0016
34	-0.119	-0.0026
35	+0.017	+0.0020
36	+0.000	+0.0045
37	+0.043	-0.0019
38	+0.059	+0.0096
39	+0.043	+0.0057
40	-0.014	+0.0029
41	-0.017	+0.0001
43	-0.014	+0.0038
44	-0.026	-0.0004
45	-0.037	+0.0039
46	+0.036	+0.0003
47	+0.032	+0.0034
48	+0.013	-0.0017
49	+0.030	+0.0002
50	-0.019	+0.0051
51	-0.048	+0.0011
52	-0.021	-0.0013
53	-0.007	+0.0034
54	-0.034	+0.0010
55	+0.082	+0.0032
56	+0.046	+0.0109
57	-0.004	-0.0042
58	-0.053	+0.0024
59	-0.022	-0.0012
60	+0.063	+0.0007
61	+0.040	+0.0175

62	-0.003	+0.0070
63	-0.021	-0.0019
64	+0.002	+0.0073
65	+0.024	-0.0019
66	-0.001	+0.0084
67	+0.019	+0.0006
68	-0.019	+0.0020
69	+0.050	+0.0030
70	-0.007	+0.0023
71	-0.020	+0.0019
72	+0.062	-0.0013
73	-0.041	+0.0005
74	-0.002	-0.0052
75	-0.037	+0.0042
76	+0.132	-0.0027
78	+0.004	+0.0068
79	-0.041	+0.0029
80	-0.042	+0.0076
100	-0.115	+0.0155
101	-0.117	-0.0044
102	-0.232	+0.0306
103	+0.041	+0.0075
104	-0.040	-0.0204
105	-0.067	+0.0056
107	-0.120	-0.0063
200	-0.032	+0.0002
201	+0.026	+0.0105
202	-0.139	-0.0216
204	-0.116	+0.0231

Number of usable observations

Flightline	Z	Roll
1	1928	1928
2	1100	1100
3	518	518
4	945	945
6	830	830
7	227	227
8	420	420
9	2741	2741
10	4494	4494
11	636	636

12	2678	2678
13	933	933
14	1764	1763
15	2514	2513
16	1888	1883
17	1987	1983
18	960	960
19	1232	1232
20	758	758
21	952	952
22	706	705
23	540	539
24	218	218
25	738	738
26	2984	2984
27	605	605
28	1685	1685
29	1656	1656
30	1814	1814
31	1085	1085
32	657	657
33	374	374
34	4626	4625
35	431	427
36	926	923
37	830	825
38	196	195
39	323	314
40	584	582
41	1005	1004
43	649	647
44	466	466
45	570	567
46	1941	1872
47	1402	1384
48	819	818
49	2266	2266
50	1039	1032
51	2431	2360
52	1530	1516
53	1591	1591
54	3994	3967
55	1343	1343

56	1837	1837
57	1261	1260
58	700	700
59	419	419
60	1589	1589
61	833	833
62	2210	2210
63	1855	1855
64	925	925
65	1859	1858
66	351	350
67	279	279
68	279	279
69	895	888
70	1576	1567
71	1648	1640
72	410	409
73	142	142
74	257	257
75	505	505
76	1635	1635
78	2573	2573
79	1807	1807
80	3437	3437
100	255	255
101	1340	1340
102	654	654
103	1342	1342
104	1525	1525
105	1730	1730
107	1172	1172
200	1563	1563
201	2092	2092
202	957	957
204	1310	1310

3.4 Resultat fra kjentflater

Det er gjort beregninger på høydeavvik mellom laserdataene og de landmålte kontrollflatene. Resultater er listet i vedlegg A.

4. LEVERANSER

4.1 Innhold i leveransene

Filformat laserdata	Beskrivelse
#.sos:	terrengpunkter på sosi-sverm. Sosi-versjon 4.0
#.las:	terreng (klasse 2) og ikke terrengpunkter (klasse 1) på lasformat
#.tif:	Geotiff grid overflate og terreng.

De nevnte laserdata er levert sammen med rapport for laserskanning, tetthets- og dekningsoversikter. Det er også levert sosi-fil med faktisk dekning av laserdata.

4.2 Oppdeling og navning av filer

Las- og sosifiler er inndelt i.h.h.t. kartbladinndeling 1:1000 (800x600m).

SOS-filer er i tillegg navnet etter følgende system:

Uklassifisert:	<kartbladindeks>_CL1.sos
Bakkepunkter:	<kartbladindeks>_CL2.sos
Støypunkter	<kartbladindeks>_CL7.sos
Punkter på bru:	<kartbladindeks>_CL10.sos

Oversiktsfiler er navnet etter følgende system:

Dekningsoversikt:	LACHNO43-45_Prosjektavgrensning.sos
Flystripe:	LACHNO43-45_Flystripe.sos
Punkttetthetskart:	LACHNO43-45_Punkttetthet-førsteretur.tif
	LACHNO43-45_Punkttetthet-terreng.tif

Dataene er lagret i følgende katalogstruktur:

- Prosjektnavn
- Rapport
- Las-data
- SOSI-Punktsky
- Dekningsoversikt og flystriper
- Tetthetsanalyse og kart
- Overflatemodell GeoTiff
- Terrengmodell GeoTiff

5.0 Vedlegg A

Kontrollflate 42

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B135_KF00042.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	607269.200	7585210.340	31.740	31.780	+0.040
2	607268.520	7585211.080	31.700	31.750	+0.050
3	607267.860	7585211.790	31.700	31.730	+0.030
4	607267.200	7585212.520	31.680	31.710	+0.030
5	607266.540	7585213.240	31.650	31.680	+0.030
6	607265.880	7585213.960	31.610	31.630	+0.020
7	607265.200	7585214.730	31.550	31.610	+0.060
8	607264.550	7585215.440	31.490	31.550	+0.060
9	607263.900	7585216.150	31.430	31.480	+0.050
10	607263.180	7585216.920	31.360	31.440	+0.080
11	607262.450	7585216.300	31.410	31.460	+0.050
12	607263.110	7585215.500	31.450	31.480	+0.030
13	607263.790	7585214.770	31.510	31.580	+0.070
14	607264.410	7585214.030	31.560	31.600	+0.040
15	607265.100	7585213.300	31.600	31.650	+0.050
16	607265.760	7585212.590	31.640	31.690	+0.050
17	607266.430	7585211.840	31.690	31.700	+0.010
18	607267.090	7585211.110	31.690	31.750	+0.060
19	607267.760	7585210.400	31.760	31.790	+0.030
20	607268.450	7585209.630	31.780	31.820	+0.040
21	607267.750	7585208.970	31.820	31.850	+0.030
22	607267.030	7585209.750	31.780	31.810	+0.030
23	607266.350	7585210.480	31.750	31.770	+0.020
24	607265.700	7585211.200	31.730	31.730	+0.000
25	607265.010	7585211.960	31.680	31.700	+0.020
26	607264.380	7585212.670	31.640	31.680	+0.040
27	607263.690	7585213.410	31.600	31.610	+0.010
28	607263.060	7585214.130	31.560	31.580	+0.020
29	607262.370	7585214.850	31.510	31.530	+0.020
30	607261.740	7585215.650	31.450	31.460	+0.010
31	607260.970	7585215.000	31.430	31.470	+0.040
32	607261.660	7585214.180	31.490	31.550	+0.060
33	607262.330	7585213.480	31.540	31.590	+0.050
34	607262.950	7585212.730	31.580	31.650	+0.070
35	607263.630	7585212.020	31.620	31.670	+0.050

36	607264.280	7585211.310	31.650	31.710	+0.060
37	607264.950	7585210.560	31.680	31.740	+0.060
38	607265.630	7585209.830	31.730	31.780	+0.050
39	607266.300	7585209.090	31.760	31.800	+0.040
40	607267.020	7585208.320	31.790	31.860	+0.070
41	607266.250	7585207.650	31.750	31.840	+0.090
42	607265.550	7585208.440	31.720	31.800	+0.080
43	607264.880	7585209.170	31.700	31.770	+0.070
44	607264.210	7585209.900	31.680	31.730	+0.050
45	607263.530	7585210.630	31.640	31.680	+0.040
46	607262.860	7585211.350	31.600	31.650	+0.050
47	607262.200	7585212.070	31.560	31.630	+0.070
48	607261.550	7585212.800	31.540	31.590	+0.050
49	607260.890	7585213.540	31.490	31.550	+0.060
50	607260.200	7585214.320	31.440	31.470	+0.030

Average dz	+0.044
Minimum dz	+0.000
Maximum dz	+0.090
Average magnitude	0.044
Root mean square	0.049
Std deviation	0.020

Kontrollflate 43_A

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B124_KF00043.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	607002.170	7586541.400	3.960	3.840	-0.120
2	607003.010	7586541.990	3.970	3.860	-0.110
3	607003.830	7586542.550	3.980	3.840	-0.140
4	607004.630	7586543.110	4.000	3.900	-0.100
5	607005.410	7586543.660	4.020	4.030	+0.010
6	607006.220	7586544.230	4.040	4.070	+0.030
7	607007.060	7586544.810	4.060	4.070	+0.010
8	607007.650	7586543.980	4.050	3.990	-0.060
9	607006.800	7586543.380	4.040	4.010	-0.030
10	607005.990	7586542.820	4.030	3.950	-0.080
11	607005.190	7586542.270	4.020	3.870	-0.150
12	607004.390	7586541.700	4.010	3.840	-0.170
13	607003.580	7586541.130	4.000	3.850	-0.150

14	607002.760	7586540.540	4.000	3.900	-0.100
15	607003.320	7586539.730	3.980	3.890	-0.090
16	607004.170	7586540.300	4.000	3.910	-0.090
17	607004.980	7586540.870	4.010	3.920	-0.090
18	607005.780	7586541.430	4.010	3.930	-0.080
19	607006.580	7586541.990	4.010	3.980	-0.030
20	607007.380	7586542.560	4.020	3.910	-0.110
21	607008.230	7586543.170	4.010	3.970	-0.040
22	607008.780	7586542.360	3.960	4.000	+0.040
23	607007.930	7586541.770	3.970	4.000	+0.030
24	607007.130	7586541.190	3.970	4.030	+0.060
25	607006.330	7586540.630	3.960	3.990	+0.030
26	607005.530	7586540.070	3.950	3.980	+0.030
27	607004.720	7586539.510	3.940	4.010	+0.070
28	607003.890	7586538.930	3.920	3.900	-0.020
29	607004.450	7586538.130	3.880	3.780	-0.100
30	607005.290	7586538.690	3.880	3.920	+0.040
31	607006.110	7586539.240	3.900	3.830	-0.070
32	607006.910	7586539.800	3.930	3.980	+0.050
33	607007.700	7586540.380	3.950	3.980	+0.030
34	607008.500	7586540.950	3.930	3.960	+0.030
35	607009.340	7586541.560	3.920	3.800	-0.120
36	607009.900	7586540.750	3.860	3.740	-0.120
37	607009.050	7586540.170	3.880	3.830	-0.050
38	607008.240	7586539.600	3.890	3.910	+0.020
39	607007.450	7586539.040	3.880	3.930	+0.050
40	607006.650	7586538.470	3.860	3.900	+0.040
41	607005.840	7586537.900	3.820	3.890	+0.070
42	607005.010	7586537.320	3.820	3.850	+0.030
43	607005.590	7586536.460	3.760	3.810	+0.050
44	607006.420	7586537.030	3.770	3.820	+0.050
45	607007.250	7586537.590	3.800	3.860	+0.060
46	607008.050	7586538.150	3.830	3.850	+0.020
47	607008.830	7586538.710	3.810	3.850	+0.040
48	607009.640	7586539.280	3.830	3.750	-0.080
49	607010.500	7586539.880	3.820	3.660	-0.160

Average dz	-0.032
Minimum dz	-0.170
Maximum dz	+0.070
Average magnitude	0.068
Root mean square	0.080
Std deviation	0.074

Kontrollflate 43_B

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B271_KF00043.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	605013.280	7567784.100	32.380	32.440	+0.060
2	605012.730	7567784.560	32.390	32.450	+0.060
3	605012.240	7567784.950	32.370	32.470	+0.100
4	605011.840	7567785.320	32.350	32.340	-0.010
5	605011.370	7567785.690	32.340	32.420	+0.080
6	605010.840	7567786.130	32.340	32.310	-0.030
7	605011.300	7567786.800	32.340	32.360	+0.020
8	605011.860	7567786.390	32.380	32.360	-0.020
9	605012.300	7567785.970	32.380	32.340	-0.040
10	605012.680	7567785.660	32.350	32.360	+0.010
11	605013.190	7567785.200	32.370	32.400	+0.030
12	605013.730	7567784.770	32.380	32.380	+0.000
13	605014.160	7567785.450	32.380	32.450	+0.070
14	605013.690	7567785.870	32.390	32.420	+0.030
15	605013.190	7567786.290	32.390	32.390	+0.000
16	605012.780	7567786.630	32.380	32.380	+0.000
17	605012.330	7567787.000	32.360	32.380	+0.020
18	605011.770	7567787.430	32.330	32.360	+0.030
19	605012.150	7567788.100	32.320	32.350	+0.030
20	605012.690	7567787.630	32.370	32.340	-0.030
21	605013.180	7567787.230	32.380	32.370	-0.010
22	605013.590	7567786.820	32.390	32.390	+0.000
23	605014.090	7567786.370	32.390	32.420	+0.030
24	605014.560	7567785.940	32.410	32.440	+0.030
25	605015.010	7567786.560	32.400	32.410	+0.010
26	605014.530	7567786.910	32.400	32.420	+0.020
27	605014.020	7567787.360	32.370	32.410	+0.040
28	605013.560	7567787.770	32.350	32.360	+0.010
29	605013.080	7567788.100	32.340	32.340	+0.000
30	605012.550	7567788.550	32.340	32.350	+0.010
31	605012.870	7567789.040	32.310	32.370	+0.060
32	605013.430	7567788.660	32.320	32.360	+0.040
33	605013.930	7567788.330	32.310	32.380	+0.070
34	605014.390	7567788.020	32.360	32.380	+0.020
35	605014.910	7567787.620	32.350	32.390	+0.040

36	605015.480	7567787.230	32.360	32.350	-0.010

Average dz	+0.021
Minimum dz	-0.040
Maximum dz	+0.100
Average magnitude	0.030
Root mean square	0.038
Std deviation	0.032

Kontrollflate 44

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B106_KF00044.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	606441.430	7587685.230	3.570	3.530	-0.040
2	606442.290	7587684.690	3.550	3.500	-0.050
3	606443.140	7587684.180	3.560	3.560	+0.000
4	606443.960	7587683.670	3.560	3.580	+0.020
5	606444.800	7587683.150	3.570	3.560	-0.010
6	606445.630	7587682.630	3.570	3.610	+0.040
7	606446.520	7587682.100	3.580	3.630	+0.050
8	606447.060	7587682.990	3.590	3.640	+0.050
9	606446.190	7587683.530	3.590	3.530	-0.060
10	606445.350	7587684.040	3.590	3.530	-0.060
11	606444.520	7587684.550	3.610	3.610	+0.000
12	606443.680	7587685.070	3.590	3.570	-0.020
13	606442.830	7587685.590	3.580	3.520	-0.060
14	606441.970	7587686.110	3.570	3.470	-0.100
15	606442.490	7587686.950	3.590	3.560	-0.030
16	606443.350	7587686.430	3.590	3.580	-0.010
17	606444.190	7587685.920	3.600	3.500	-0.100
18	606445.030	7587685.420	3.620	3.590	-0.030
19	606445.870	7587684.900	3.600	3.620	+0.020
20	606446.710	7587684.380	3.600	3.590	-0.010
21	606447.590	7587683.830	3.600	3.610	+0.010
22	606448.100	7587684.670	3.620	3.610	-0.010
23	606447.200	7587685.200	3.620	3.600	-0.020
24	606446.370	7587685.720	3.620	3.580	-0.040
25	606445.530	7587686.230	3.630	3.550	-0.080
26	606444.690	7587686.730	3.620	3.550	-0.070
27	606443.850	7587687.260	3.600	3.520	-0.080

28	606442.990	7587687.800	3.590	3.520	-0.070
29	606443.500	7587688.620	3.600	3.610	+0.010
30	606444.370	7587688.110	3.620	3.590	-0.030
31	606445.220	7587687.600	3.630	3.610	-0.020
32	606446.060	7587687.110	3.640	3.710	+0.070
33	606446.890	7587686.590	3.650	3.640	-0.010
34	606447.730	7587686.060	3.640	3.600	-0.040
35	606448.610	7587685.510	3.640	3.720	+0.080
36	606449.120	7587686.350	3.670	3.620	-0.050
37	606448.250	7587686.900	3.670	3.670	+0.000
38	606447.410	7587687.430	3.680	3.750	+0.070
39	606446.590	7587687.950	3.680	3.690	+0.010
40	606445.740	7587688.440	3.660	3.590	-0.070
41	606444.880	7587688.950	3.640	3.590	-0.050
42	606444.010	7587689.470	3.630	3.630	+0.000
43	606444.550	7587690.340	3.650	3.690	+0.040
44	606445.410	7587689.800	3.670	3.640	-0.030
45	606446.260	7587689.290	3.690	3.600	-0.090
46	606447.100	7587688.790	3.700	3.680	-0.020
47	606447.930	7587688.270	3.700	3.700	+0.000
48	606448.770	7587687.740	3.700	3.750	+0.050
49	606449.650	7587687.220	3.700	3.680	-0.020

Average dz	-0.018
Minimum dz	-0.100
Maximum dz	+0.080
Average magnitude	0.039
Root mean square	0.048
Std deviation	0.045

Kontrollflate 48

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B018_KF00048.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	606181.020	7606472.070	49.160	49.170	+0.010
2	606180.650	7606471.080	49.170	49.200	+0.030
3	606180.220	7606470.100	49.170	49.310	+0.140
4	606179.800	7606469.050	49.200	49.280	+0.080
5	606179.350	7606467.940	49.220	49.270	+0.050
6	606178.880	7606466.590	49.250	49.410	+0.160

7	606179.910	7606466.320	49.260	49.230	-0.030
8	606180.350	7606467.550	49.220	49.320	+0.100
9	606180.670	7606468.670	49.200	49.330	+0.130
10	606181.100	7606469.690	49.190	49.170	-0.020
11	606181.470	7606470.600	49.210	49.160	-0.050
12	606181.850	7606471.610	49.210	49.150	-0.060
13	606182.790	7606471.340	49.190	49.180	-0.010
14	606182.440	7606470.300	49.200	49.160	-0.040
15	606182.130	7606469.370	49.240	49.170	-0.070
16	606181.810	7606468.450	49.250	49.210	-0.040
17	606181.410	7606467.290	49.260	49.250	-0.010
18	606181.030	7606466.070	49.290	49.390	+0.100
19	606182.060	7606465.830	49.290	49.310	+0.020
20	606182.340	7606466.970	49.270	49.240	-0.030
21	606182.660	7606468.100	49.240	49.210	-0.030
22	606182.930	7606468.990	49.220	49.190	-0.030
23	606183.200	7606469.890	49.180	49.240	+0.060
24	606183.630	7606470.950	49.160	49.280	+0.120
25	606184.570	7606470.590	49.150	49.150	+0.000
26	606184.240	7606469.460	49.190	49.240	+0.050
27	606183.970	7606468.570	49.200	49.330	+0.130
28	606183.760	7606467.730	49.240	49.270	+0.030
29	606183.490	7606466.630	49.270	49.260	-0.010
30	606183.160	7606465.560	49.270	49.350	+0.080
31	606184.430	7606465.250	49.270	49.360	+0.090
32	606184.610	7606466.300	49.250	49.350	+0.100
33	606184.820	7606467.330	49.230	49.350	+0.120
34	606185.020	7606468.250	49.220	49.350	+0.130
35	606185.240	7606469.130	49.180	49.330	+0.150
36	606185.520	7606470.180	49.160	49.280	+0.120

Average dz	+0.044
Minimum dz	-0.070
Maximum dz	+0.160
Average magnitude	0.068
Root mean square	0.082
Std deviation	0.070

Kontrollflate 49

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B013_KF00049.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	607709.170	7607392.240	89.130	89.180	+0.050
2	607709.090	7607393.240	89.160	89.110	-0.050
3	607709.040	7607394.170	89.190	89.110	-0.080
4	607708.970	7607395.080	89.180	89.110	-0.070
5	607708.890	7607395.960	89.170	89.140	-0.030
6	607708.020	7607395.980	89.180	89.160	-0.020
7	607708.050	7607395.030	89.160	89.100	-0.060
8	607708.090	7607394.210	89.160	89.080	-0.080
9	607708.120	7607393.250	89.130	89.080	-0.050
10	607708.140	7607392.190	89.100	89.210	+0.110
11	607707.150	7607392.090	89.090	89.160	+0.070
12	607707.120	7607393.180	89.130	89.070	-0.060
13	607707.100	7607394.180	89.160	89.110	-0.050
14	607707.070	7607394.980	89.160	89.100	-0.060
15	607707.030	7607395.980	89.140	89.220	+0.080
16	607706.360	7607395.970	89.150	89.160	+0.010
17	607706.310	7607394.980	89.130	89.090	-0.040
18	607706.310	7607394.170	89.150	89.120	-0.030
19	607706.240	7607393.130	89.140	89.110	-0.030
20	607706.240	7607392.030	89.090	89.200	+0.110
21	607705.210	7607391.900	89.090	89.180	+0.090
22	607705.200	7607393.090	89.110	89.080	-0.030
23	607705.150	7607394.040	89.140	89.100	-0.040
24	607705.240	7607394.930	89.140	89.100	-0.040
25	607705.230	7607395.910	89.130	89.180	+0.050
26	607704.350	7607395.890	89.150	89.200	+0.050
27	607704.320	7607394.940	89.140	89.100	-0.040
28	607704.280	7607394.010	89.150	89.080	-0.070
29	607704.280	7607393.000	89.120	89.060	-0.060
30	607704.270	7607391.840	89.080	89.170	+0.090
31	607703.290	7607391.690	89.070	89.110	+0.040
32	607703.340	7607392.960	89.130	89.060	-0.070
33	607703.340	7607393.890	89.160	89.090	-0.070
34	607703.400	7607394.860	89.130	89.060	-0.070
35	607703.450	7607395.860	89.150	89.160	+0.010

Average dz	-0.013
Minimum dz	-0.080
Maximum dz	+0.110
Average magnitude	0.056
Root mean square	0.061

Std deviation 0.061

Kontrollflate 50

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B012_KF00050.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	606693.240	7607174.960	83.140	83.200	+0.060
2	606692.830	7607175.910	83.170	83.120	-0.050
3	606692.430	7607176.750	83.190	83.090	-0.100
4	606692.110	7607177.560	83.180	83.180	+0.000
5	606691.740	7607178.420	83.180	83.200	+0.020
6	606691.400	7607179.360	83.170	83.250	+0.080
7	606690.700	7607179.110	83.180	83.130	-0.050
8	606690.970	7607178.190	83.180	83.140	-0.040
9	606691.250	7607177.340	83.180	83.160	-0.020
10	606691.590	7607176.420	83.180	83.120	-0.060
11	606691.900	7607175.570	83.170	83.110	-0.060
12	606692.230	7607174.520	83.130	83.230	+0.100
13	606691.210	7607174.060	83.130	83.230	+0.100
14	606690.810	7607175.090	83.160	83.140	-0.020
15	606690.510	7607176.000	83.180	83.140	-0.040
16	606690.230	7607176.920	83.200	83.130	-0.070
17	606689.950	7607177.760	83.200	83.110	-0.090
18	606689.690	7607178.770	83.170	83.200	+0.030
19	606688.880	7607178.640	83.180	83.180	+0.000
20	606689.200	7607177.560	83.200	83.120	-0.080
21	606689.430	7607176.720	83.200	83.140	-0.060
22	606689.660	7607175.800	83.210	83.140	-0.070
23	606689.970	7607174.820	83.170	83.140	-0.030
24	606690.350	7607173.760	83.140	83.230	+0.090
25	606689.470	7607173.400	83.140	83.210	+0.070
26	606689.120	7607174.450	83.190	83.190	+0.000
27	606688.870	7607175.460	83.200	83.140	-0.060
28	606688.620	7607176.390	83.210	83.170	-0.040
29	606688.380	7607177.260	83.210	83.170	-0.040
30	606688.100	7607178.310	83.200	83.180	-0.020
31	606687.280	7607178.050	83.200	83.120	-0.080
32	606687.490	7607177.010	83.210	83.150	-0.060
33	606687.690	7607176.090	83.230	83.130	-0.100
34	606687.930	7607175.180	83.200	83.120	-0.080

35	606688.200	7607174.180	83.200	83.120	-0.080
36	606688.540	7607173.020	83.150	83.200	+0.050

Average dz	-0.022
Minimum dz	-0.100
Maximum dz	+0.100
Average magnitude	0.056
Root mean square	0.063
Std deviation	0.060

Kontrollflate 53

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B022_KF00053.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	607480.030	7598251.220	16.870	16.870	+0.000
2	607479.160	7598251.260	16.890	16.920	+0.030
3	607478.330	7598251.280	16.940	16.960	+0.020
4	607477.540	7598251.300	16.990	17.000	+0.010
5	607476.730	7598251.410	17.040	17.060	+0.020
6	607475.860	7598251.500	17.100	17.090	-0.010
7	607475.090	7598251.560	17.140	17.140	+0.000
8	607475.110	7598252.290	17.180	17.170	-0.010
9	607475.890	7598252.250	17.130	17.160	+0.030
10	607476.700	7598252.200	17.080	17.140	+0.060
11	607477.510	7598252.130	17.030	17.070	+0.040
12	607478.340	7598252.170	16.990	17.030	+0.040
13	607479.150	7598252.120	16.950	16.940	-0.010
14	607480.050	7598252.180	16.900	16.900	+0.000
15	607480.110	7598252.930	16.920	16.950	+0.030
16	607479.240	7598253.050	16.970	16.970	+0.000
17	607478.480	7598253.080	17.030	17.040	+0.010
18	607477.590	7598253.170	17.080	17.080	+0.000
19	607476.830	7598253.220	17.130	17.130	+0.000
20	607475.990	7598253.270	17.180	17.210	+0.030
21	607475.170	7598253.300	17.220	17.230	+0.010
22	607475.220	7598254.220	17.250	17.260	+0.010
23	607476.040	7598254.170	17.200	17.190	-0.010
24	607476.860	7598254.130	17.150	17.150	+0.000
25	607477.630	7598254.090	17.100	17.090	-0.010
26	607478.520	7598254.000	17.050	17.020	-0.030

27	607479.220	7598253.880	17.010	16.970	-0.040
28	607480.110	7598253.870	16.930	16.960	+0.030
29	607480.200	7598254.610	16.930	16.910	-0.020
30	607479.340	7598254.660	16.990	16.960	-0.030
31	607478.510	7598254.730	17.040	16.990	-0.050
32	607477.750	7598254.680	17.080	17.080	+0.000
33	607476.970	7598254.770	17.130	17.130	+0.000
34	607476.100	7598254.820	17.180	17.180	+0.000
35	607475.190	7598254.890	17.250	17.250	+0.000
36	607477.600	7598253.150	17.080	17.080	+0.000

Average dz	+0.004
Minimum dz	-0.050
Maximum dz	+0.060
Average magnitude	0.016
Root mean square	0.023
Std deviation	0.023

Kontrollflate 54

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B029_KF00054.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	608975.080	7593576.950	175.680	175.760	+0.080
2	608975.990	7593576.260	175.680	175.780	+0.100
3	608976.660	7593575.420	175.670	175.800	+0.130
4	608977.190	7593574.910	175.660	175.760	+0.100
5	608977.790	7593574.240	175.660	175.720	+0.060
6	608978.330	7593573.630	175.680	175.710	+0.030
7	608979.000	7593572.950	175.660	175.790	+0.130
8	608978.270	7593572.200	175.740	175.830	+0.090
9	608977.670	7593572.830	175.750	175.880	+0.130
10	608977.130	7593573.480	175.770	175.800	+0.030
11	608976.510	7593574.130	175.780	175.840	+0.060
12	608975.970	7593574.670	175.770	175.880	+0.110
13	608975.350	7593575.420	175.740	175.860	+0.120
14	608974.550	7593576.120	175.730	175.850	+0.120
15	608974.050	7593575.460	175.770	175.910	+0.140
16	608974.760	7593574.720	175.800	175.910	+0.110
17	608975.380	7593574.150	175.830	175.920	+0.090
18	608975.970	7593573.650	175.800	175.900	+0.100

19	608976.530	7593572.920	175.830	175.910	+0.080
20	608977.040	7593572.350	175.820	175.920	+0.100
21	608977.720	7593571.700	175.780	175.880	+0.100
22	608977.190	7593571.200	175.830	175.900	+0.070
23	608976.530	7593571.780	175.850	175.930	+0.080
24	608975.990	7593572.290	175.860	175.980	+0.120
25	608975.360	7593572.930	175.870	175.980	+0.110
26	608974.770	7593573.480	175.860	175.970	+0.110
27	608974.230	7593574.050	175.860	175.970	+0.110
28	608973.580	7593574.830	175.830	175.940	+0.110
29	608973.090	7593574.250	175.890	175.990	+0.100
30	608973.750	7593573.580	175.920	176.020	+0.100
31	608974.270	7593573.100	175.930	176.020	+0.090
32	608974.860	7593572.450	175.930	176.050	+0.120
33	608975.430	7593571.800	175.930	176.020	+0.090
34	608976.040	7593571.330	175.910	176.010	+0.100
35	608976.620	7593570.700	175.930	176.000	+0.070
36	608975.980	7593570.000	175.970	176.050	+0.080
37	608975.460	7593570.630	175.980	176.090	+0.110
38	608974.950	7593571.150	176.010	176.080	+0.070
39	608974.380	7593571.780	176.010	176.100	+0.090
40	608973.790	7593572.400	176.020	176.060	+0.040
41	608973.200	7593572.980	175.990	176.100	+0.110
42	608972.580	7593573.610	176.000	176.050	+0.050

Average dz	+0.094
Minimum dz	+0.030
Maximum dz	+0.140
Average magnitude	0.094
Root mean square	0.097
Std deviation	0.027

Kontrollflate 55

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B031_KF00055.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	610053.280	7593638.830	42.700	42.650	-0.050
2	610052.230	7593639.000	42.690	42.720	+0.030
3	610051.280	7593639.180	42.800	42.760	-0.040
4	610050.460	7593639.340	42.840	42.820	-0.020

5	610049.670	7593639.470	42.840	42.860	+0.020
6	610048.940	7593639.580	42.900	42.890	-0.010
7	610048.200	7593639.620	42.880	42.920	+0.040
8	610048.050	7593638.690	42.990	42.940	-0.050
9	610048.760	7593638.630	42.940	42.900	-0.040
10	610049.440	7593638.510	42.910	42.890	-0.020
11	610050.250	7593638.350	42.870	42.840	-0.030
12	610051.020	7593638.360	42.850	42.810	-0.040
13	610051.940	7593638.140	42.800	42.720	-0.080
14	610052.990	7593637.860	42.740	42.670	-0.070
15	610052.830	7593637.010	42.790	42.690	-0.100
16	610051.820	7593637.160	42.830	42.750	-0.080
17	610050.900	7593637.320	42.850	42.820	-0.030
18	610050.140	7593637.420	42.890	42.850	-0.040
19	610049.300	7593637.560	42.930	42.880	-0.050
20	610048.630	7593637.670	42.960	42.890	-0.070
21	610047.910	7593637.750	42.990	42.940	-0.050
22	610047.740	7593636.620	43.000	42.960	-0.040
23	610048.490	7593636.490	42.970	42.890	-0.080
24	610049.140	7593636.430	42.930	42.910	-0.020
25	610049.920	7593636.380	42.910	42.860	-0.050
26	610050.720	7593636.240	42.870	42.810	-0.060
27	610051.610	7593636.180	42.860	42.800	-0.060
28	610052.660	7593636.060	42.780	42.700	-0.080
29	610052.580	7593635.330	42.810	42.710	-0.100
30	610051.570	7593635.350	42.850	42.790	-0.060
31	610050.650	7593635.470	42.900	42.850	-0.050
32	610049.880	7593635.510	42.890	42.900	+0.010
33	610049.090	7593635.610	42.950	42.900	-0.050
34	610048.400	7593635.700	42.980	42.920	-0.060
35	610047.660	7593635.750	43.020	42.960	-0.060

Average dz	-0.044
Minimum dz	-0.100
Maximum dz	+0.040
Average magnitude	0.050
Root mean square	0.055
Std deviation	0.033

Kontrollflate 56

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B031_KF00056.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	610904.030	7593842.660	39.830	39.790	-0.040
2	610902.520	7593842.850	39.870	39.850	-0.020
3	610901.660	7593843.000	39.880	39.890	+0.010
4	610900.700	7593843.120	39.900	39.880	-0.020
5	610899.560	7593843.450	39.940	39.860	-0.080
6	610898.650	7593843.670	39.950	39.890	-0.060
7	610898.920	7593844.840	39.950	39.910	-0.040
8	610899.820	7593844.570	39.930	39.870	-0.060
9	610900.800	7593844.370	39.910	39.900	-0.010
10	610901.800	7593844.160	39.900	39.870	-0.030
11	610902.630	7593844.090	39.880	39.850	-0.030
12	610904.000	7593843.810	39.850	39.790	-0.060
13	610904.140	7593844.750	39.850	39.810	-0.040
14	610902.870	7593845.110	39.860	39.800	-0.060
15	610902.080	7593845.220	39.880	39.810	-0.070
16	610901.010	7593845.450	39.910	39.890	-0.020
17	610900.120	7593845.710	39.940	39.890	-0.050
18	610899.310	7593846.090	39.940	39.940	+0.000
19	610899.750	7593847.290	39.920	39.880	-0.040
20	610900.480	7593846.940	39.900	39.880	-0.020
21	610901.160	7593846.630	39.880	39.880	+0.000
22	610902.240	7593846.280	39.860	39.780	-0.080
23	610902.960	7593846.180	39.840	39.770	-0.070
24	610904.230	7593845.730	39.830	39.810	-0.020
25	610904.480	7593846.600	39.780	39.790	+0.010
26	610903.330	7593847.020	39.810	39.760	-0.050
27	610902.550	7593847.280	39.830	39.750	-0.080
28	610901.600	7593847.830	39.860	39.810	-0.050
29	610900.970	7593848.170	39.870	39.870	+0.000
30	610900.270	7593848.490	39.900	39.850	-0.050
31	610900.660	7593849.350	39.860	39.840	-0.020
32	610901.400	7593849.090	39.830	39.800	-0.030
33	610902.030	7593848.720	39.810	39.760	-0.050
34	610902.780	7593848.380	39.800	39.740	-0.060
35	610903.650	7593848.030	39.760	39.720	-0.040
36	610904.680	7593847.630	39.750	39.730	-0.020

Average dz -0.037

Minimum dz -0.080

Maximum dz	+0.010
Average magnitude	0.039
Root mean square	0.045
Std deviation	0.025

Kontrollflate 57

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B054_KF00057.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	600511.710	7590342.290	15.740	15.700	-0.040
2	600511.390	7590343.250	15.720	15.680	-0.040
3	600511.100	7590344.110	15.720	outside	*
4	600510.820	7590344.970	15.730	15.720	-0.010
5	600510.600	7590345.730	15.740	15.720	-0.020
6	600510.320	7590346.550	15.760	15.730	-0.030
7	600511.140	7590346.760	15.810	15.770	-0.040
8	600511.320	7590345.990	15.790	15.760	-0.030
9	600511.480	7590345.210	15.770	15.760	-0.010
10	600511.760	7590344.450	15.770	outside	*
11	600512.060	7590343.490	15.770	outside	*
12	600512.330	7590342.520	15.780	15.740	-0.040
13	600512.940	7590342.750	15.810	15.920	+0.110
14	600512.700	7590343.710	15.800	outside	*
15	600512.440	7590344.620	15.800	15.820	+0.020
16	600512.220	7590345.370	15.820	15.800	-0.020
17	600512.070	7590346.210	15.830	15.790	-0.040
18	600511.880	7590346.930	15.840	15.820	-0.020
19	600512.520	7590347.110	15.880	15.850	-0.030
20	600512.720	7590346.410	15.880	15.840	-0.040
21	600512.900	7590345.600	15.850	15.830	-0.020
22	600513.100	7590344.900	15.850	15.800	-0.050
23	600513.390	7590343.980	15.840	15.930	+0.090
24	600513.600	7590343.070	15.840	15.950	+0.110
25	600514.220	7590343.280	15.880	16.000	+0.120
26	600513.990	7590344.120	15.880	15.950	+0.070
27	600513.780	7590345.060	15.900	15.870	-0.030
28	600513.610	7590345.800	15.920	15.850	-0.070
29	600513.430	7590346.620	15.940	15.870	-0.070
30	600513.240	7590347.280	15.920	15.890	-0.030
31	600514.170	7590347.510	15.980	15.950	-0.030

32	600514.320	7590346.860	15.970	15.940	-0.030
33	600514.520	7590346.080	15.970	15.930	-0.040
34	600514.640	7590345.360	15.960	15.900	-0.060
35	600514.880	7590344.470	15.940	16.000	+0.060
36	600515.120	7590343.590	15.920	16.050	+0.130

Average dz	-0.004
Minimum dz	-0.070
Maximum dz	+0.130
Average magnitude	0.048
Root mean square	0.058
Std deviation	0.059

Kontrollflate 58

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B111_KF00058.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	586601.660	7586915.920	8.910	8.870	-0.040
2	586601.350	7586916.400	8.950	8.890	-0.060
3	586600.980	7586916.870	8.970	8.910	-0.060
4	586600.660	7586917.340	8.990	8.940	-0.050
5	586600.390	7586917.720	9.010	8.960	-0.050
6	586600.090	7586918.170	9.030	8.980	-0.050
7	586599.630	7586917.830	9.040	9.000	-0.040
8	586599.940	7586917.430	9.010	8.980	-0.030
9	586600.190	7586917.040	8.950	8.950	+0.000
10	586600.450	7586916.580	8.980	8.920	-0.060
11	586600.760	7586916.030	8.920	8.920	+0.000
12	586601.060	7586915.540	8.900	8.900	+0.000
13	586600.540	7586915.200	8.900	8.870	-0.030
14	586600.260	7586915.650	8.930	8.880	-0.050
15	586599.930	7586916.190	8.970	8.900	-0.070
16	586599.650	7586916.630	8.990	8.930	-0.060
17	586599.380	7586917.030	9.000	8.960	-0.040
18	586599.100	7586917.460	9.040	9.000	-0.040
19	586598.630	7586917.160	9.040	9.010	-0.030
20	586598.910	7586916.750	9.000	8.960	-0.040
21	586599.170	7586916.360	8.980	8.930	-0.050
22	586599.450	7586915.880	8.960	8.920	-0.040
23	586599.760	7586915.390	8.950	8.890	-0.060

24	586600.010	7586914.900	8.920	8.870	-0.050
25	586599.540	7586914.580	8.920	8.880	-0.040
26	586599.250	7586915.060	8.950	8.900	-0.050
27	586598.940	7586915.560	8.970	8.930	-0.040
28	586598.660	7586916.000	8.990	9.020	+0.030
29	586598.390	7586916.430	9.010	8.940	-0.070
30	586598.110	7586916.910	9.040	8.950	-0.090
31	586597.450	7586916.540	9.050	8.990	-0.060
32	586597.790	7586916.050	9.030	8.970	-0.060
33	586598.060	7586915.670	9.010	8.950	-0.060
34	586598.330	7586915.240	8.990	8.920	-0.070
35	586598.640	7586914.710	8.950	8.930	-0.020
36	586598.960	7586914.190	8.930	8.930	+0.000

Average dz	-0.042
Minimum dz	-0.090
Maximum dz	+0.030
Average magnitude	0.044
Root mean square	0.049
Std deviation	0.024

Kontrollflate 59

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B130_KF00059.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	592590.690	7585963.140	35.790	35.890	+0.100
2	592590.790	7585962.450	35.840	35.830	-0.010
3	592590.710	7585961.700	35.830	35.820	-0.010
4	592590.660	7585961.050	35.840	35.870	+0.030
5	592590.620	7585960.510	35.830	35.890	+0.060
6	592590.640	7585959.960	35.860	35.910	+0.050
7	592591.560	7585959.880	35.950	35.910	-0.040
8	592591.520	7585960.480	35.930	35.870	-0.060
9	592591.510	7585961.060	35.910	35.880	-0.030
10	592591.520	7585961.660	35.890	35.890	+0.000
11	592591.640	7585962.430	35.910	35.880	-0.030
12	592591.520	7585963.160	35.890	35.870	-0.020
13	592592.430	7585963.160	35.930	35.900	-0.030
14	592592.570	7585962.430	35.950	35.930	-0.020
15	592592.380	7585961.680	35.960	35.950	-0.010

16	592592.390	7585961.050	36.020	35.950	-0.070
17	592592.430	7585960.450	35.990	35.980	-0.010
18	592592.540	7585959.830	36.030	36.000	-0.030
19	592593.420	7585959.660	36.080	36.050	-0.030
20	592593.400	7585960.360	36.050	36.030	-0.020
21	592593.310	7585960.970	36.040	36.020	-0.020
22	592593.270	7585961.540	36.010	36.010	+0.000
23	592593.470	7585962.380	36.020	36.000	-0.020
24	592593.330	7585963.140	36.000	35.960	-0.040
25	592594.200	7585963.150	36.060	36.020	-0.040
26	592594.410	7585962.420	36.070	36.010	-0.060
27	592594.300	7585961.600	36.070	36.050	-0.020
28	592594.270	7585960.940	36.080	36.050	-0.030
29	592594.260	7585960.310	36.080	36.070	-0.010
30	592594.260	7585959.540	36.120	36.090	-0.030
31	592595.030	7585959.380	36.140	36.180	+0.040
32	592595.050	7585960.110	36.140	36.200	+0.060
33	592595.110	7585960.920	36.120	36.170	+0.050
34	592595.150	7585961.580	36.100	36.150	+0.050
35	592595.150	7585962.360	36.100	36.140	+0.040
36	592595.190	7585963.150	36.110	36.130	+0.020
37	592591.530	7585961.020	35.930	35.880	-0.050

Average dz	-0.006
Minimum dz	-0.070
Maximum dz	+0.100
Average magnitude	0.034
Root mean square	0.040
Std deviation	0.040

Kontrollflate 60

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B177_KF00060.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	598975.840	7575421.910	2.890	2.870	-0.020
2	598974.890	7575422.420	2.880	2.960	+0.080
3	598974.020	7575422.890	2.880	2.880	+0.000
4	598973.150	7575423.360	2.870	2.850	-0.020
5	598972.290	7575423.830	2.870	2.880	+0.010
6	598971.430	7575424.300	2.870	2.930	+0.060

7	598970.540	7575424.800	2.870	2.830	-0.040
8	598969.710	7575425.230	2.870	2.880	+0.010
9	598968.840	7575425.710	2.870	2.850	-0.020
10	598967.920	7575426.200	2.880	2.890	+0.010
11	598968.410	7575427.100	2.910	2.960	+0.050
12	598969.330	7575426.600	2.900	2.960	+0.060
13	598970.200	7575426.110	2.900	2.920	+0.020
14	598971.050	7575425.680	2.900	2.890	-0.010
15	598971.940	7575425.200	2.900	2.860	-0.040
16	598972.790	7575424.710	2.900	2.870	-0.030
17	598973.670	7575424.240	2.900	2.950	+0.050
18	598974.530	7575423.770	2.920	2.960	+0.040
19	598975.400	7575423.300	2.910	2.950	+0.040
20	598976.330	7575422.790	2.930	2.950	+0.020
21	598976.800	7575423.650	2.970	2.990	+0.020
22	598975.880	7575424.140	2.950	2.950	+0.000
23	598975.010	7575424.610	2.960	2.930	-0.030
24	598974.150	7575425.080	2.940	2.930	-0.010
25	598973.270	7575425.550	2.930	2.930	+0.000
26	598972.420	7575426.050	2.920	2.860	-0.060
27	598971.540	7575426.530	2.920	2.920	+0.000
28	598970.660	7575426.950	2.940	2.970	+0.030
29	598969.810	7575427.430	2.940	2.970	+0.030
30	598968.860	7575427.950	2.940	2.910	-0.030
31	598969.340	7575428.820	2.920	2.900	-0.020
32	598970.280	7575428.310	2.920	2.960	+0.040
33	598971.140	7575427.820	2.930	2.950	+0.020
34	598972.020	7575427.390	2.940	2.970	+0.030
35	598972.900	7575426.900	2.930	3.010	+0.080
36	598973.750	7575426.410	2.930	2.940	+0.010
37	598974.620	7575425.950	2.930	2.930	+0.000
38	598975.490	7575425.480	2.930	2.980	+0.050
39	598976.370	7575425.000	2.930	3.000	+0.070
40	598977.280	7575424.510	2.940	3.030	+0.090
41	598977.770	7575425.420	2.940	3.010	+0.070
42	598976.870	7575425.920	2.930	2.980	+0.050
43	598975.990	7575426.390	2.930	3.000	+0.070
44	598975.140	7575426.860	2.920	2.930	+0.010
45	598974.260	7575427.330	2.910	2.910	+0.000
46	598973.400	7575427.800	2.900	2.990	+0.090
47	598972.530	7575428.280	2.900	2.970	+0.070
48	598971.660	7575428.750	2.900	2.980	+0.080
49	598970.790	7575429.210	2.900	2.950	+0.050

50	598969.840	7575429.720	2.890	2.910	+0.020

Average dz	+0.022
Minimum dz	-0.060
Maximum dz	+0.090
Average magnitude	0.035
Root mean square	0.044
Std deviation	0.038

Kontrollflate 61

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B194_KF00061.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	599557.010	7573643.010	11.090	11.090	+0.000
2	599557.770	7573643.680	11.130	11.090	-0.040
3	599558.510	7573644.340	11.140	11.130	-0.010
4	599559.240	7573644.990	11.160	11.140	-0.020
5	599559.970	7573645.640	11.150	11.130	-0.020
6	599560.700	7573646.300	11.150	11.160	+0.010
7	599561.510	7573647.010	11.150	11.080	-0.070
8	599560.780	7573647.810	11.130	11.180	+0.050
9	599560.030	7573647.110	11.140	11.140	+0.000
10	599559.290	7573646.450	11.140	11.130	-0.010
11	599558.560	7573645.790	11.140	11.140	+0.000
12	599557.830	7573645.150	11.140	11.090	-0.050
13	599557.100	7573644.500	11.120	11.100	-0.020
14	599556.300	7573643.770	11.070	11.070	+0.000
15	599555.660	7573644.510	11.070	11.070	+0.000
16	599556.440	7573645.240	11.100	11.080	-0.020
17	599557.180	7573645.890	11.110	11.160	+0.050
18	599557.910	7573646.550	11.120	11.110	-0.010
19	599558.630	7573647.200	11.140	11.200	+0.060
20	599559.370	7573647.850	11.130	11.120	-0.010
21	599560.110	7573648.530	11.110	11.150	+0.040
22	599559.460	7573649.250	11.150	11.140	-0.010
23	599558.710	7573648.570	11.140	11.140	+0.000
24	599557.980	7573647.910	11.130	11.130	+0.000
25	599557.260	7573647.260	11.130	11.090	-0.040
26	599556.540	7573646.590	11.120	11.200	+0.080
27	599555.800	7573645.940	11.120	11.110	-0.010

28	599555.010	7573645.240	11.100	11.100	+0.000
29	599554.370	7573645.970	11.120	11.150	+0.030
30	599555.150	7573646.680	11.120	11.160	+0.040
31	599555.890	7573647.330	11.130	11.220	+0.090
32	599556.620	7573647.980	11.130	11.180	+0.050
33	599557.330	7573648.640	11.140	11.160	+0.020
34	599558.070	7573649.300	11.150	11.140	-0.010
35	599558.810	7573649.970	11.140	11.170	+0.030
36	599558.160	7573650.710	11.160	11.120	-0.040
37	599557.430	7573650.030	11.140	11.140	+0.000
38	599556.690	7573649.370	11.140	11.150	+0.010
39	599555.960	7573648.720	11.130	11.100	-0.030
40	599555.230	7573648.070	11.130	11.140	+0.010
41	599554.500	7573647.410	11.130	11.120	-0.010
42	599553.720	7573646.700	11.120	11.070	-0.050
43	599553.000	7573647.500	11.130	11.090	-0.040
44	599553.770	7573648.210	11.140	11.150	+0.010
45	599554.510	7573648.860	11.150	11.130	-0.020
46	599555.250	7573649.520	11.150	11.170	+0.020
47	599555.970	7573650.150	11.150	11.140	-0.010
48	599556.720	7573650.800	11.160	11.200	+0.040
49	599557.480	7573651.460	11.160	11.160	+0.000

Average dz	+0.002
Minimum dz	-0.070
Maximum dz	+0.090
Average magnitude	0.024
Root mean square	0.033
Std deviation	0.033

Kontrollflate 62

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B211_KF00062.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	599875.990	7571647.160	5.100	5.000	-0.100
2	599875.080	7571646.560	5.100	5.090	-0.010
3	599874.260	7571646.010	5.100	5.070	-0.030
4	599873.450	7571645.470	5.100	5.010	-0.090
5	599872.640	7571644.920	5.090	5.010	-0.080
6	599871.820	7571644.380	5.090	5.010	-0.080

7	599870.970	7571643.810	5.050	5.020	-0.030
8	599871.540	7571642.980	5.050	4.980	-0.070
9	599872.440	7571643.580	5.070	5.000	-0.070
10	599873.270	7571644.120	5.090	5.030	-0.060
11	599874.080	7571644.660	5.080	5.060	-0.020
12	599874.910	7571645.180	5.080	5.070	-0.010
13	599875.700	7571645.750	5.090	5.040	-0.050
14	599876.560	7571646.320	5.090	5.040	-0.050
15	599877.100	7571645.510	5.090	5.040	-0.050
16	599876.240	7571644.910	5.100	5.050	-0.050
17	599875.420	7571644.360	5.080	5.070	-0.010
18	599874.610	7571643.830	5.080	5.040	-0.040
19	599873.800	7571643.300	5.080	5.000	-0.080
20	599872.970	7571642.760	5.060	5.010	-0.050
21	599872.080	7571642.170	5.060	5.000	-0.060
22	599872.620	7571641.370	5.060	5.010	-0.050
23	599873.530	7571641.970	5.050	4.980	-0.070
24	599874.360	7571642.500	5.070	5.020	-0.050
25	599875.180	7571643.040	5.070	5.080	+0.010
26	599875.980	7571643.580	5.060	5.050	-0.010
27	599876.810	7571644.120	5.080	5.030	-0.050
28	599877.630	7571644.690	5.090	5.050	-0.040
29	599878.180	7571643.890	5.090	5.060	-0.030
30	599877.340	7571643.320	5.070	5.040	-0.030
31	599876.520	7571642.790	5.060	5.030	-0.030
32	599875.720	7571642.240	5.050	5.060	+0.010
33	599874.900	7571641.700	5.050	5.000	-0.050
34	599874.080	7571641.170	5.050	4.970	-0.080
35	599873.170	7571640.550	5.040	5.030	-0.010
36	599873.730	7571639.730	5.010	4.990	-0.020
37	599874.630	7571640.340	5.030	4.970	-0.060
38	599875.460	7571640.870	5.050	4.990	-0.060
39	599876.280	7571641.410	5.060	5.030	-0.030
40	599877.090	7571641.940	5.080	5.030	-0.050
41	599877.910	7571642.500	5.090	5.040	-0.050
42	599878.740	7571643.080	5.100	5.040	-0.060
43	599879.350	7571642.180	5.110	5.070	-0.040
44	599878.500	7571641.610	5.080	5.060	-0.020
45	599877.670	7571641.080	5.080	5.020	-0.060
46	599876.860	7571640.550	5.050	5.000	-0.050
47	599876.030	7571640.020	5.030	5.010	-0.020
48	599875.220	7571639.480	5.030	5.000	-0.030
49	599874.330	7571638.860	4.990	4.930	-0.060

Average dz	-0.044
Minimum dz	-0.100
Maximum dz	+0.010
Average magnitude	0.045
Root mean square	0.051
Std deviation	0.025

Kontrollflate 63

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B103_KF00063.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	596459.240	7587425.010	143.000	143.070	+0.070
2	596459.060	7587424.320	143.060	143.110	+0.050
3	596458.890	7587423.660	143.090	143.130	+0.040
4	596458.720	7587422.970	143.110	143.150	+0.040
5	596458.530	7587422.180	143.140	143.200	+0.060
6	596458.250	7587421.670	143.180	143.270	+0.090
7	596459.040	7587421.290	143.200	143.210	+0.010
8	596459.320	7587421.830	143.170	143.180	+0.010
9	596459.490	7587422.580	143.140	143.160	+0.020
10	596459.790	7587423.220	143.110	143.070	-0.040
11	596459.930	7587423.860	143.050	143.050	+0.000
12	596460.120	7587424.530	143.010	143.060	+0.050
13	596461.000	7587424.100	143.030	142.970	-0.060
14	596460.800	7587423.490	143.080	143.010	-0.070
15	596460.690	7587422.920	143.120	143.070	-0.050
16	596460.400	7587422.260	143.150	143.140	-0.010
17	596460.230	7587421.510	143.180	143.190	+0.010
18	596459.970	7587420.960	143.210	143.270	+0.060
19	596460.770	7587420.710	143.200	143.280	+0.080
20	596460.980	7587421.280	143.160	143.280	+0.120
21	596461.130	7587421.910	143.130	143.200	+0.070
22	596461.500	7587422.560	143.100	143.110	+0.010
23	596461.650	7587423.100	143.070	143.040	-0.030
24	596461.830	7587423.630	143.030	143.060	+0.030
25	596462.700	7587423.250	143.040	143.160	+0.120
26	596462.510	7587422.750	143.080	143.160	+0.080
27	596462.290	7587422.240	143.100	143.190	+0.090
28	596462.000	7587421.650	143.140	143.240	+0.100

29	596461.780	7587421.070	143.170	143.230	+0.060
30	596461.570	7587420.460	143.170	143.260	+0.090
31	596462.450	7587420.220	143.170	143.280	+0.110
32	596462.570	7587420.830	143.150	143.260	+0.110
33	596462.740	7587421.320	143.120	143.230	+0.110
34	596462.950	7587421.910	143.100	143.180	+0.080
35	596463.120	7587422.460	143.070	143.150	+0.080
36	596463.240	7587422.980	143.040	143.120	+0.080

Average dz	+0.046
Minimum dz	-0.070
Maximum dz	+0.120
Average magnitude	0.061
Root mean square	0.070
Std deviation	0.053

Kontrollflate 122

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B008_KF00122.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	602734.700	7607461.650	282.780	282.720	-0.060
2	602735.220	7607462.370	282.770	282.770	+0.000
3	602735.760	7607463.150	282.830	282.840	+0.010
4	602736.330	7607463.930	282.900	282.910	+0.010
5	602736.850	7607464.700	282.990	282.980	-0.010
6	602737.330	7607465.430	283.020	283.030	+0.010
7	602737.830	7607466.140	283.140	283.090	-0.050
8	602738.370	7607466.910	283.130	283.130	+0.000
9	602738.920	7607467.670	283.250	283.230	-0.020
10	602739.390	7607468.430	283.320	283.290	-0.030
11	602739.870	7607469.180	283.310	283.370	+0.060
12	602740.450	7607470.030	283.370	283.400	+0.030
13	602739.650	7607470.570	283.390	283.380	-0.010
14	602739.060	7607469.690	283.330	283.360	+0.030
15	602738.560	7607468.980	283.220	283.270	+0.050
16	602738.090	7607468.250	283.170	283.180	+0.010
17	602737.580	7607467.550	283.110	283.100	-0.010
18	602737.050	7607466.860	283.070	283.060	-0.010
19	602736.540	7607465.960	283.010	283.040	+0.030
20	602736.080	7607465.290	282.940	282.970	+0.030

21	602735.610	7607464.530	282.900	282.920	+0.020
22	602735.060	7607463.720	282.850	282.850	+0.000
23	602734.510	7607462.990	282.790	282.780	-0.010
24	602734.030	7607462.310	282.730	282.720	-0.010
25	602733.310	7607462.770	282.710	282.470	-0.240
26	602733.750	7607463.510	282.780	outside	*
27	602734.260	7607464.250	282.840	outside	*
28	602734.770	7607465.060	282.910	282.990	+0.080
29	602735.260	7607465.820	282.950	283.010	+0.060
30	602735.700	7607466.520	283.030	283.080	+0.050
31	602736.270	7607467.410	283.100	283.170	+0.070
32	602736.760	7607468.080	283.160	283.280	+0.120
33	602737.340	7607468.790	283.170	283.260	+0.090
34	602737.810	7607469.480	283.250	283.330	+0.080
35	602738.230	7607470.120	283.330	283.440	+0.110
36	602738.830	7607471.120	283.400	283.510	+0.110

Average dz	+0.018
Minimum dz	-0.240
Maximum dz	+0.120
Average magnitude	0.045
Root mean square	0.066
Std deviation	0.064

Kontrollflate 124

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B009_KF00124.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	603858.840	7607414.960	37.870	37.820	-0.050
2	603859.310	7607415.820	37.900	37.790	-0.110
3	603859.760	7607416.640	37.890	37.800	-0.090
4	603860.220	7607417.410	37.910	37.780	-0.130
5	603860.690	7607418.160	37.940	37.820	-0.120
6	603861.320	7607419.140	37.940	37.820	-0.120
7	603861.930	7607419.960	37.930	37.800	-0.130
8	603862.470	7607420.740	37.950	37.790	-0.160
9	603861.470	7607421.460	38.030	37.910	-0.120
10	603860.870	7607420.730	38.030	37.890	-0.140
11	603860.380	7607419.820	38.040	37.880	-0.160
12	603859.710	7607418.970	38.000	37.890	-0.110

13	603859.160	7607418.280	38.000	37.870	-0.130
14	603858.670	7607417.410	37.980	37.860	-0.120
15	603858.180	7607416.570	37.960	37.850	-0.110
16	603857.230	7607417.260	38.010	37.900	-0.110
17	603857.800	7607418.050	38.020	37.910	-0.110
18	603858.350	7607418.840	38.040	37.930	-0.110
19	603858.900	7607419.570	38.040	37.950	-0.090
20	603859.560	7607420.360	38.050	37.960	-0.090
21	603860.100	7607421.230	38.050	37.950	-0.100
22	603860.580	7607422.040	38.070	38.030	-0.040
23	603859.730	7607422.570	38.110	38.050	-0.060
24	603859.260	7607421.760	38.120	38.100	-0.020
25	603858.780	7607420.950	38.100	38.030	-0.070
26	603858.110	7607420.220	38.090	37.990	-0.100
27	603857.480	7607419.410	38.050	37.930	-0.120
28	603856.970	7607418.760	38.030	37.960	-0.070
29	603856.430	7607417.930	38.010	37.890	-0.120
30	603855.590	7607418.510	38.000	38.010	+0.010
31	603856.160	7607419.310	38.050	38.000	-0.050
32	603856.630	7607420.020	38.110	37.970	-0.140
33	603857.200	7607420.800	38.140	38.040	-0.100
34	603857.880	7607421.610	38.170	38.120	-0.050
35	603858.350	7607422.340	38.180	38.210	+0.030
36	603858.860	7607423.180	38.170	38.090	-0.080
37	603858.840	7607414.960	37.870	37.820	-0.050
38	603859.310	7607415.820	37.900	37.790	-0.110
39	603859.760	7607416.640	37.890	37.800	-0.090
40	603860.220	7607417.410	37.910	37.780	-0.130
41	603860.690	7607418.160	37.940	37.820	-0.120
42	603861.320	7607419.140	37.940	37.820	-0.120
43	603861.930	7607419.960	37.930	37.800	-0.130
44	603862.470	7607420.740	37.950	37.790	-0.160
45	603861.470	7607421.460	38.030	37.910	-0.120
46	603860.870	7607420.730	38.030	37.890	-0.140
47	603860.380	7607419.820	38.040	37.880	-0.160
48	603859.710	7607418.970	38.000	37.890	-0.110
49	603859.160	7607418.280	38.000	37.870	-0.130
50	603858.670	7607417.410	37.980	37.860	-0.120
51	603858.180	7607416.570	37.960	37.850	-0.110
52	603857.230	7607417.260	38.010	37.900	-0.110
53	603857.800	7607418.050	38.020	37.910	-0.110
54	603858.350	7607418.840	38.040	37.930	-0.110
55	603858.900	7607419.570	38.040	37.950	-0.090

56	603859.560	7607420.360	38.050	37.960	-0.090
57	603860.100	7607421.230	38.050	37.950	-0.100
58	603860.580	7607422.040	38.070	38.030	-0.040
59	603859.730	7607422.570	38.110	38.050	-0.060
60	603859.260	7607421.760	38.120	38.100	-0.020
61	603858.780	7607420.950	38.100	38.030	-0.070
62	603858.110	7607420.220	38.090	37.990	-0.100
63	603857.480	7607419.410	38.050	37.930	-0.120
64	603856.970	7607418.760	38.030	37.960	-0.070
65	603856.430	7607417.930	38.010	37.890	-0.120
66	603855.590	7607418.510	38.000	38.010	+0.010
67	603856.160	7607419.310	38.050	38.000	-0.050
68	603856.630	7607420.020	38.110	37.970	-0.140
69	603857.200	7607420.800	38.140	38.040	-0.100
70	603857.880	7607421.610	38.170	38.120	-0.050
71	603858.350	7607422.340	38.180	38.210	+0.030
72	603858.860	7607423.180	38.170	38.090	-0.080

Average dz	-0.094
Minimum dz	-0.160
Maximum dz	+0.030
Average magnitude	0.096
Root mean square	0.103
Std deviation	0.043

Kontrollflate 125

H:\04_Laser_Projects_2014\40445_Nordland_2pt\diverse\Controlpoints\CP_Block\B009_KF00125.txt

Number	Easting	Northing	Known Z	Laser Z	Dz
1	603814.190	7607731.360	92.700	92.770	+0.070
2	603813.330	7607731.030	92.660	92.660	+0.000
3	603812.420	7607730.570	92.550	92.600	+0.050
4	603811.470	7607730.000	92.440	92.460	+0.020
5	603810.480	7607729.530	92.360	92.440	+0.080
6	603809.640	7607729.090	92.270	92.300	+0.030
7	603808.760	7607728.640	92.200	92.200	+0.000
8	603807.910	7607728.110	92.080	92.180	+0.100
9	603807.110	7607727.570	92.000	92.030	+0.030
10	603806.350	7607727.070	91.880	91.990	+0.110
11	603805.550	7607726.580	91.840	91.920	+0.080

12	603804.630	7607726.150	91.760	91.820	+0.060
13	603804.160	7607727.010	91.740	91.800	+0.060
14	603805.060	7607727.490	91.840	91.860	+0.020
15	603805.920	7607727.980	91.980	92.010	+0.030
16	603806.660	7607728.380	92.040	92.060	+0.020
17	603807.460	7607728.880	92.110	92.120	+0.010
18	603808.370	7607729.360	92.170	92.220	+0.050
19	603809.230	7607729.820	92.280	92.320	+0.040
20	603810.040	7607730.250	92.360	92.380	+0.020
21	603810.990	7607730.740	92.440	92.480	+0.040
22	603812.040	7607731.300	92.570	92.620	+0.050
23	603812.920	7607731.820	92.680	92.700	+0.020
24	603813.870	7607732.180	92.780	92.770	-0.010
25	603813.330	7607733.100	92.800	outside	*
26	603812.370	7607732.680	92.680	92.800	+0.120
27	603811.560	7607732.170	92.580	92.680	+0.100
28	603810.350	7607731.580	92.440	92.530	+0.090
29	603809.550	7607731.140	92.340	92.420	+0.080
30	603808.760	7607730.750	92.280	92.420	+0.140
31	603807.990	7607730.330	92.200	92.310	+0.110
32	603807.010	7607729.840	92.100	92.220	+0.120
33	603806.200	7607729.290	92.030	92.150	+0.120
34	603805.460	7607728.910	91.960	92.140	+0.180
35	603804.610	7607728.500	91.870	92.020	+0.150
36	603803.760	7607728.000	91.770	91.830	+0.060
37	603814.190	7607731.360	92.700	92.770	+0.070
38	603813.330	7607731.030	92.660	92.660	+0.000
39	603812.420	7607730.570	92.550	92.600	+0.050
40	603811.470	7607730.000	92.440	92.460	+0.020
41	603810.480	7607729.530	92.360	92.440	+0.080
42	603809.640	7607729.090	92.270	92.300	+0.030
43	603808.760	7607728.640	92.200	92.200	+0.000
44	603807.910	7607728.110	92.080	92.180	+0.100
45	603807.110	7607727.570	92.000	92.030	+0.030
46	603806.350	7607727.070	91.880	91.990	+0.110
47	603805.550	7607726.580	91.840	91.920	+0.080
48	603804.630	7607726.150	91.760	91.820	+0.060
49	603804.160	7607727.010	91.740	91.800	+0.060
50	603805.060	7607727.490	91.840	91.860	+0.020
51	603805.920	7607727.980	91.980	92.010	+0.030
52	603806.660	7607728.380	92.040	92.060	+0.020
53	603807.460	7607728.880	92.110	92.120	+0.010
54	603808.370	7607729.360	92.170	92.220	+0.050

55	603809.230	7607729.820	92.280	92.320	+0.040
56	603810.040	7607730.250	92.360	92.380	+0.020
57	603810.990	7607730.740	92.440	92.480	+0.040
58	603812.040	7607731.300	92.570	92.620	+0.050
59	603812.920	7607731.820	92.680	92.700	+0.020
60	603813.870	7607732.180	92.780	92.770	-0.010
61	603813.330	7607733.100	92.800	outside	*
62	603812.370	7607732.680	92.680	92.800	+0.120
63	603811.560	7607732.170	92.580	92.680	+0.100
64	603810.350	7607731.580	92.440	92.530	+0.090
65	603809.550	7607731.140	92.340	92.420	+0.080
66	603808.760	7607730.750	92.280	92.420	+0.140
67	603807.990	7607730.330	92.200	92.310	+0.110
68	603807.010	7607729.840	92.100	92.220	+0.120
69	603806.200	7607729.290	92.030	92.150	+0.120
70	603805.460	7607728.910	91.960	92.140	+0.180
71	603804.610	7607728.500	91.870	92.020	+0.150
72	603803.760	7607728.000	91.770	91.830	+0.060

Average dz	+0.064
Minimum dz	-0.010
Maximum dz	+0.180
Average magnitude	0.065
Root mean square	0.079
Std deviation	0.047